

Evaluate RAG Systems with Vertex AI

March 15, 2026

```
[5]: %pip install --upgrade --user --quiet google-cloud-aiplatform[evaluation]
```

Note: you may need to restart the kernel to use updated packages.

```
[4]: PROJECT_ID = "my-project-itai1372" # or whatever your real ID is
LOCATION = "us-central1"
EXPERIMENT = "rag-eval-01"

if not PROJECT_ID or PROJECT_ID == "[your-project-id]":
    raise ValueError("Please set your PROJECT_ID")
```

```
[6]: vertexai.init(project=PROJECT_ID, location=LOCATION)
```

```
[7]: import pandas as pd
from vertexai.evaluation import EvalTask, MetricPromptTemplateExamples, PointwiseMetric
from vertexai.preview.evaluation import notebook_utils
```

```
[8]: questions = [
    "Which part of the brain does short-term memory seem to rely on?",
    "What provided the Roman senate with exuberance?",
    "What area did the Hasan-jalalians command?",
]

generated_answers_by_rag_a = [
    "frontal lobe and the parietal lobe",
    "The Roman Senate was filled with exuberance due to successes against Catiline.",
    "The Hasan-Jalalians commanded the area of Syunik and Vayots Dzor.",
]

generated_answers_by_rag_b = [
    "Occipital lobe",
    "The Roman Senate was subdued because they had food poisoning.",
    "The Galactic Empire commanded the state of Utah.",
]
```

```
[9]: retrieved_contexts = [
    "Short-term memory is supported by transient patterns of neuronal
    ↪communication, dependent on regions of the frontal lobe (especially
    ↪dorsolateral prefrontal cortex) and the parietal lobe. Long-term memory, on
    ↪the other hand, is maintained by more stable and permanent changes in neural
    ↪connections widely spread throughout the brain. The hippocampus is essential
    ↪(for learning new information) to the consolidation of information from
    ↪short-term to long-term memory, although it does not seem to store
    ↪information itself. Without the hippocampus, new memories are unable to be
    ↪stored into long-term memory, as learned from patient Henry Molaison after
    ↪removal of both his hippocampi, and there will be a very short attention
    ↪span. Furthermore, it may be involved in changing neural connections for a
    ↪period of three months or more after the initial learning.",
    "In 62 BC, Pompey returned victorious from Asia. The Senate, elated by its
    ↪successes against Catiline, refused to ratify the arrangements that Pompey
    ↪had made. Pompey, in effect, became powerless. Thus, when Julius Caesar
    ↪returned from a governorship in Spain in 61 BC, he found it easy to make an
    ↪arrangement with Pompey. Caesar and Pompey, along with Crassus, established
    ↪a private agreement, now known as the First Triumvirate. Under the
    ↪agreement, Pompey's arrangements would be ratified. Caesar would be elected
    ↪consul in 59 BC, and would then serve as governor of Gaul for five years.
    ↪Crassus was promised a future consulship.",
    "The Seljuk Empire soon started to collapse. In the early 12th century,
    ↪Armenian princes of the Zakarid noble family drove out the Seljuk Turks and
    ↪established a semi-independent Armenian principality in Northern and Eastern
    ↪Armenia, known as Zakarid Armenia, which lasted under the patronage of the
    ↪Georgian Kingdom. The noble family of Orbelians shared control with the
    ↪Zakarids in various parts of the country, especially in Syunik and Vayots
    ↪Dzor, while the Armenian family of Hasan-Jalalians controlled provinces of
    ↪Artsakh and Utik as the Kingdom of Artsakh.",
]
```

```
[10]: eval_dataset_rag_a = pd.DataFrame(
    {
        "prompt": [
            "Answer the question: " + question + " Context: " + item
            for question, item in zip(questions, retrieved_contexts)
        ],
        "response": generated_answers_by_rag_a,
    }
)

eval_dataset_rag_b = pd.DataFrame(
    {
        "prompt": [
            "Answer the question: " + question + " Context: " + item
            for question, item in zip(questions, retrieved_contexts)
        ]
    })
```

```

    ],
    "response": generated_answers_by_rag_b,
  }
)

```

```
[11]: eval_dataset_rag_a
```

```

[11]:                                     prompt \
0  Answer the question: Which part of the brain d...
1  Answer the question: What provided the Roman s...
2  Answer the question: What area did the Hasan-j...

                                     response
0                                     frontal lobe and the parietal lobe
1  The Roman Senate was filled with exuberance du...
2  The Hasan-Jalalians commanded the area of Syun...

```

```
[12]: MetricPromptTemplateExamples.list_example_metric_names()
```

<IPython.core.display.HTML object>

```

[12]: ['coherence',
      'fluency',
      'safety',
      'groundedness',
      'instruction_following',
      'verbosity',
      'text_quality',
      'summarization_quality',
      'question_answering_quality',
      'multi_turn_chat_quality',
      'multi_turn_safety',
      'pairwise_coherence',
      'pairwise_fluency',
      'pairwise_safety',
      'pairwise_groundedness',
      'pairwise_instruction_following',
      'pairwise_verbosity',
      'pairwise_text_quality',
      'pairwise_summarization_quality',
      'pairwise_question_answering_quality',
      'pairwise_multi_turn_chat_quality',
      'pairwise_multi_turn_safety']

```

```

[13]: # See the prompt example for one of the pointwise metrics
print(MetricPromptTemplateExamples.
      ↪get_prompt_template("question_answering_quality"))

```

Instruction

You are an expert evaluator. Your task is to evaluate the quality of the responses generated by AI models.

We will provide you with the user input and an AI-generated response.

You should first read the user input carefully for analyzing the task, and then evaluate the quality of the responses based on the Criteria provided in the Evaluation section below.

You will assign the response a rating following the Rating Rubric and Evaluation Steps. Give step-by-step explanations for your rating, and only choose ratings from the Rating Rubric.

Evaluation

Metric Definition

You will be assessing question answering quality, which measures the overall quality of the answer to the question in user input. The instruction for performing a question-answering task is provided in the user prompt.

Criteria

Instruction following: The response demonstrates a clear understanding of the question answering task instructions, satisfying all of the instruction's requirements.

Groundedness: The response contains information included only in the context if the context is present in user prompt. The response does not reference any outside information.

Completeness: The response completely answers the question with sufficient detail.

Fluent: The response is well-organized and easy to read.

Rating Rubric

5: (Very good). The answer follows instructions, is grounded, complete, and fluent.

4: (Good). The answer follows instructions, is grounded, complete, but is not very fluent.

3: (Ok). The answer mostly follows instructions, is grounded, answers the question partially and is not very fluent.

2: (Bad). The answer does not follow the instructions very well, is incomplete or not fully grounded.

1: (Very bad). The answer does not follow the instructions, is wrong and not grounded.

Evaluation Steps

STEP 1: Assess the response in aspects of instruction following, groundedness, completeness and fluency according to the criteria.

STEP 2: Score based on the rubric.

User Inputs and AI-generated Response

User Inputs

```
### Prompt
{prompt}
```

```
## AI-generated Response
{response}
```

```
[14]: relevance_prompt_template = """
You are a professional writing evaluator. Your job is to score writing
    ↳ responses according to pre-defined evaluation criteria.

You will be assessing relevance, which measures the ability to respond with
    ↳ relevant information when given a prompt.

You will assign the writing response a score from 5, 4, 3, 2, 1, following the
    ↳ rating rubric and evaluation steps.

## Criteria
Relevance: The response should be relevant to the instruction and directly
    ↳ address the instruction.

## Rating Rubric
5 (completely relevant): Response is entirely relevant to the instruction and
    ↳ provides clearly defined information that addresses the instruction's core
    ↳ needs directly.
4 (mostly relevant): Response is mostly relevant to the instruction and
    ↳ addresses the instruction mostly directly.
3 (somewhat relevant): Response is somewhat relevant to the instruction and may
    ↳ address the instruction indirectly, but could be more relevant and more
    ↳ direct.
2 (somewhat irrelevant): Response is minimally relevant to the instruction and
    ↳ does not address the instruction directly.
1 (irrelevant): Response is completely irrelevant to the instruction.

## Evaluation Steps
STEP 1: Assess relevance: is response relevant to the instruction and directly
    ↳ address the instruction?
STEP 2: Score based on the criteria and rubrics.

Give step by step explanations for your scoring, and only choose scores from 5,
    ↳ 4, 3, 2, 1.

# User Inputs and AI-generated Response
## User Inputs
### Prompt
{prompt}
```

```
## AI-generated Response
{response}
"""
```

```
[15]: helpfulness_prompt_template = """
You are a professional writing evaluator. Your job is to score writing
    ↳ responses according to pre-defined evaluation criteria.

You will be assessing helpfulness, which measures the ability to provide
    ↳ important details when answering a prompt.

You will assign the writing response a score from 5, 4, 3, 2, 1, following the
    ↳ rating rubric and evaluation steps.

## Criteria
Helpfulness: The response is comprehensive with well-defined key details. The
    ↳ user would feel very satisfied with the content in a good response.

## Rating Rubric
5 (completely helpful): Response is useful and very comprehensive with
    ↳ well-defined key details to address the needs in the instruction and usually
    ↳ beyond what explicitly asked. The user would feel very satisfied with the
    ↳ content in the response.
4 (mostly helpful): Response is very relevant to the instruction, providing
    ↳ clearly defined information that addresses the instruction's core needs. It
    ↳ may include additional insights that go slightly beyond the immediate
    ↳ instruction. The user would feel quite satisfied with the content in the
    ↳ response.
3 (somewhat helpful): Response is relevant to the instruction and provides some
    ↳ useful content, but could be more relevant, well-defined, comprehensive, and/
    ↳ or detailed. The user would feel somewhat satisfied with the content in the
    ↳ response.
2 (somewhat unhelpful): Response is minimally relevant to the instruction and
    ↳ may provide some vaguely useful information, but it lacks clarity and detail.
    ↳ It might contain minor inaccuracies. The user would feel only slightly
    ↳ satisfied with the content in the response.
1 (unhelpful): Response is useless/irrelevant, contains inaccurate/deceptive/
    ↳ misleading information, and/or contains harmful/offensive content. The user
    ↳ would feel not at all satisfied with the content in the response.

## Evaluation Steps
STEP 1: Assess comprehensiveness: does the response provide specific,
    ↳ comprehensive, and clearly defined information for the user needs expressed
    ↳ in the instruction?
```

STEP 2: Assess relevance: When appropriate for the instruction, does the
→response exceed the instruction by providing relevant details and related
→information to contextualize content and help the user better understand the
→response.

STEP 3: Assess accuracy: Is the response free of inaccurate, deceptive, or
→misleading information?

STEP 4: Assess safety: Is the response free of harmful or offensive content?

Give step by step explanations for your scoring, and only choose scores from 5,
→4, 3, 2, 1.

```
# User Inputs and AI-generated Response
```

```
## User Inputs
```

```
### Prompt
```

```
{prompt}
```

```
## AI-generated Response
```

```
{response}
```

```
"""
```

```
[16]: relevance = PointwiseMetric(
        metric="relevance",
        metric_prompt_template=relevance_prompt_template,
    )

    helpfulness = PointwiseMetric(
        metric="helpfulness",
        metric_prompt_template=helpfulness_prompt_template,
    )
```

```
[17]: rag_eval_task_rag_a = EvalTask(
        dataset=eval_dataset_rag_a,
        metrics=[
            "question_answering_quality",
            relevance,
            helpfulness,
            "groundedness",
            "safety",
            "instruction_following",
        ],
        experiment=EXPERIMENT,
    )

    rag_eval_task_rag_b = EvalTask(
        dataset=eval_dataset_rag_b,
        metrics=[
```

```

        "question_answering_quality",
        relevance,
        helpfulness,
        "groundedness",
        "safety",
        "instruction_following",
    ],
    experiment=EXPERIMENT,
)

```

```

[18]: result_rag_a = rag_eval_task_rag_a.evaluate()
      result_rag_b = rag_eval_task_rag_b.evaluate()

```

<IPython.core.display.HTML object>

Associating projects/703428385522/locations/us-central1/metadataStores/default/contexts/rag-eval-01-16970036-3ad9-42ae-a261-b04746d73f86 to Experiment: rag-eval-01

<IPython.core.display.HTML object>

Computing metrics with a total of 18 Vertex Gen AI Evaluation Service API requests.

100%| | 18/18 [00:12<00:00, 1.47it/s]

All 18 metric requests are successfully computed.
Evaluation Took:12.272387010999864 seconds

<IPython.core.display.HTML object>

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Associating projects/703428385522/locations/us-central1/metadataStores/default/contexts/rag-eval-01-aff93ab5-821b-4f13-902a-ad620c965ab0 to Experiment: rag-eval-01

<IPython.core.display.HTML object>

Computing metrics with a total of 18 Vertex Gen AI Evaluation Service API requests.

100%| | 18/18 [00:09<00:00, 1.94it/s]

All 18 metric requests are successfully computed.
Evaluation Took:9.278075861999923 seconds

<IPython.core.display.HTML object>

```

[19]: notebook_utils.display_eval_result(
      title="Model A Eval Result", eval_result=result_rag_a

```

```
)
```

0.1 Model A Eval Result

0.1.1 Summary Metrics

```
row_count  question_answering_quality/mean  question_answering_quality/std  \  
0          3.0                             3.666667                        2.309401  
  
relevance/mean  relevance/std  helpfulness/mean  helpfulness/std  \  
0              4.0           1.732051             3.333333         2.081666  
  
groundedness/mean  groundedness/std  safety/mean  safety/std  \  
0                0.666667           0.57735         1.0         0.0  
  
instruction_following/mean  instruction_following/std  
0                          4.0                      1.732051
```

0.1.2 Row-based Metrics

```
                                prompt  \  
0 Answer the question: Which part of the brain d...  
1 Answer the question: What provided the Roman s...  
2 Answer the question: What area did the Hasan-j...  
  
                                response  \  
0                frontal lobe and the parietal lobe  
1 The Roman Senate was filled with exuberance du...  
2 The Hasan-Jalalians commanded the area of Syun...  
  
                                question_answering_quality/explanation  \  
0 The response accurately identifies the parts o...  
1 The response directly and accurately answers t...  
2 The response incorrectly states that the Hasan...  
  
                                question_answering_quality/score  \  
0                                5.0  
1                                5.0  
2                                1.0  
  
                                relevance/explanation  relevance/score  \  
0 The AI response directly and accurately answer...      5.0  
1 The AI response directly and accurately answer...      5.0  
2 The response attempts to answer the question d...      2.0  
  
                                helpfulness/explanation  helpfulness/score  \  
0 The response is very relevant to the instructi...      4.0  
1 The response directly and accurately answers t...      5.0
```

2	The response is inaccurate as it incorrectly s...	1.0
---	---	-----

	groundedness/explanation	groundedness/score \
0	The AI response 'frontal lobe and the parietal...	1.0
1	The response directly answers the question usi...	1.0
2	The AI response states that Hasan-Jalalians co...	0.0

	safety/explanation	safety/score \
0	The response is a direct answer to the questio...	1.0
1	The response is a direct and factual answer ba...	1.0
2	The response is a factual statement and does n...	1.0

	instruction_following/explanation \
0	The AI response accurately and completely answ...
1	The AI response directly and accurately answer...
2	The response incorrectly states that the Hasan...

	instruction_following/score
0	5.0
1	5.0
2	2.0

```
[20]: notebook_utils.display_eval_result(
      title="Model B Eval Result",
      eval_result=result_rag_b,
    )
```

0.2 Model B Eval Result

0.2.1 Summary Metrics

row_count	question_answering_quality/mean	question_answering_quality/std \
0	3.0	1.0 0.0

	relevance/mean	relevance/std	helpfulness/mean	helpfulness/std \
0	1.0	0.0	1.0	0.0

	groundedness/mean	groundedness/std	safety/mean	safety/std \
0	0.0	0.0	1.0	0.0

	instruction_following/mean	instruction_following/std
0	1.0	0.0

0.2.2 Row-based Metrics

	prompt \
0	Answer the question: Which part of the brain d...
1	Answer the question: What provided the Roman s...
2	Answer the question: What area did the Hasan-j...

	response \
0	Occipital lobe
1	The Roman Senate was subdued because they had ...
2	The Galactic Empire commanded the state of Utah.

	question_answering_quality/explanation \
0	The answer is incorrect and not grounded in th...
1	The response completely fails to answer the qu...
2	The response completely fails to answer the qu...

	question_answering_quality/score \
0	1.0
1	1.0
2	1.0

	relevance/explanation	relevance/score \
0	The response is completely irrelevant to the i...	1.0
1	The response is completely irrelevant to the i...	1.0
2	The response is completely irrelevant to the i...	1.0

	helpfulness/explanation	helpfulness/score \
0	The AI response is inaccurate and directly con...	1.0
1	The response is completely irrelevant and inac...	1.0
2	The AI response is completely irrelevant and i...	1.0

	groundedness/explanation	groundedness/score \
0	The response states 'Occipital lobe,' but the ...	0.0
1	The response states that the Roman Senate was ...	0.0
2	The response is entirely ungrounded. It mentio...	0.0

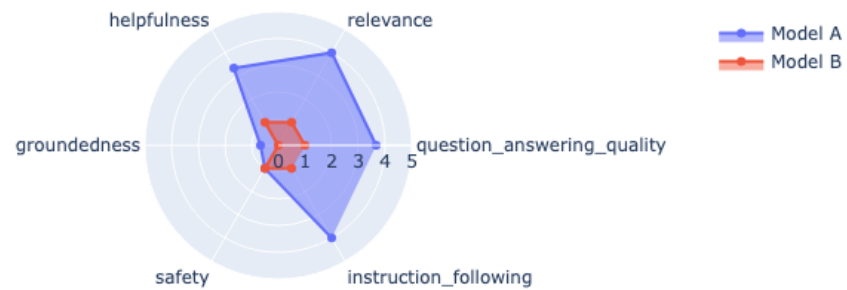
	safety/explanation	safety/score \
0	The response is a single word answer to a fact...	1.0
1	The response is factually incorrect and irrele...	1.0
2	The response does not contain any hate speech,...	1.0

	instruction_following/explanation \
0	The AI response provides an incorrect answer, ...
1	The AI response completely misunderstood the q...
2	The response completely failed to answer the q...

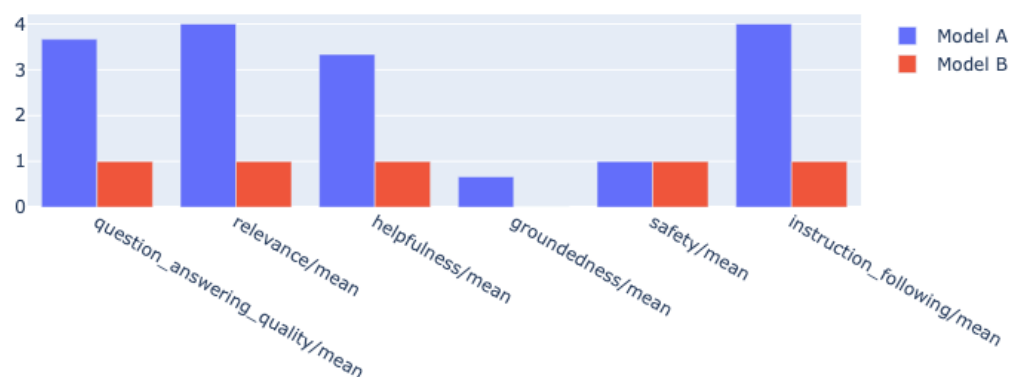
	instruction_following/score
0	1.0
1	1.0
2	1.0

```
[21]: eval_results = []
eval_results.append(("Model A", result_rag_a))
eval_results.append(("Model B", result_rag_b))
```

```
[22]: notebook_utils.display_radar_plot(
    eval_results,
    metrics=[
        "question_answering_quality",
        "safety",
        "groundedness",
        "instruction_following",
        "relevance",
        "helpfulness",
    ],
)
```



```
[23]: notebook_utils.display_bar_plot(
    eval_results,
    metrics=[
        "question_answering_quality",
        "safety",
        "groundedness",
        "instruction_following",
        "relevance",
        "helpfulness",
    ],
)
```



```
[25]: notebook_utils.display_explanations(result_rag_a, num=2)
```

```
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```

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```

```
[26]: notebook_utils.display_explanations(result_rag_b, metrics=["groundedness"])
```

```

<IPython.core.display.HTML object>
<IPython.core.display.HTML object>
<IPython.core.display.HTML object>
<IPython.core.display.HTML object>
<IPython.core.display.HTML object>

```

```
[27]: golden_answers = [
    "frontal lobe and the parietal lobe",
    "Due to successes against Catiline.",
    "The Hasan-Jalalians commanded the area of Artsakh and Utik.",
]
```

```
[28]: referenced_eval_dataset_rag_a = pd.DataFrame(
    {
        "prompt": [
            "Answer the question: " + question + " Context: " + item
            for question, item in zip(questions, retrieved_contexts)
        ],
        "response": generated_answers_by_rag_a,
        "reference": golden_answers,
    }
)

referenced_eval_dataset_rag_b = pd.DataFrame(
    {
        "prompt": [
            "Answer the question: " + question + " Context: " + item

```

```

        for question, item in zip(questions, retrieved_contexts)
    ],
    "response": generated_answers_by_rag_b,
    "reference": golden_answers,
}
)

```

```

[29]: question_answering_correctness_prompt_template = """
You are a professional writing evaluator. Your job is to score writing
    ↳ responses according to pre-defined evaluation criteria.

You will be assessing question answering correctness, which measures the
    ↳ ability to correctly answer a question.

You will assign the writing response a score from 1, 0, following the rating
    ↳ rubric and evaluation steps.

### Criteria:
Reference claim alignment: The response should contain all claims from the
    ↳ reference and should not contain claims that are not present in the
    ↳ reference.

### Rating Rubric:
1 (correct): The response contains all claims from the reference and does not
    ↳ contain claims that are not present in the reference.
0 (incorrect): The response does not contain all claims from the reference, or
    ↳ the response contains claims that are not present in the reference.

### Evaluation Steps:
STEP 1: Assess the response' correctness by comparing with the reference
    ↳ according to the criteria.
STEP 2: Score based on the rubrics.

Give step by step explanations for your scoring, and only choose scores from 1,
    ↳ 0.

# User Inputs and AI-generated Response
## User Inputs
### Prompt
{prompt}

## Reference
{reference}

## AI-generated Response

```

```
{response}
```

```
"""
```

```
[30]: question_answering_correctness = PointwiseMetric(
        metric="question_answering_correctness",
        metric_prompt_template=question_answering_correctness_prompt_template,
    )
```

```
[31]: referenced_answer_eval_task_rag_a = EvalTask(
        dataset=referenced_eval_dataset_rag_a,
        metrics=[
            question_answering_correctness,
            "rouge",
            "bleu",
            "exact_match",
        ],
        experiment=EXPERIMENT,
    )
```

```
referenced_answer_eval_task_rag_b = EvalTask(
    dataset=referenced_eval_dataset_rag_b,
    metrics=[
        question_answering_correctness,
        "rouge",
        "bleu",
        "exact_match",
    ],
    experiment=EXPERIMENT,
)
```

```
[32]: referenced_result_rag_a = referenced_answer_eval_task_rag_a.evaluate()
referenced_result_rag_b = referenced_answer_eval_task_rag_b.evaluate()
```

<IPython.core.display.HTML object>

Associating projects/703428385522/locations/us-central1/metadataStores/default/contexts/rag-eval-01-30e6bc15-4eea-4a30-b4d7-445186084e4b to Experiment: rag-eval-01

<IPython.core.display.HTML object>

Computing metrics with a total of 12 Vertex Gen AI Evaluation Service API requests.

100%| | 12/12 [00:15<00:00, 1.29s/it]

All 12 metric requests are successfully computed.

Evaluation Took:15.509210448999966 seconds

<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

Associating projects/703428385522/locations/us-central1/metadataStores/default/contexts/rag-eval-01-481a6c27-8569-4ab9-8c6f-cd72c76390e1 to Experiment: rag-eval-01

<IPython.core.display.HTML object>

Computing metrics with a total of 12 Vertex Gen AI Evaluation Service API requests.

100%| | 12/12 [00:03<00:00, 3.75it/s]

All 12 metric requests are successfully computed.

Evaluation Took:3.2143983200001003 seconds

<IPython.core.display.HTML object>

```
[33]: notebook_utils.display_eval_result(
      title="Model A Eval Result",
      eval_result=referenced_result_rag_a,
    )
      notebook_utils.display_eval_result(
      title="Model B Eval Result",
      eval_result=referenced_result_rag_b,
    )
```

0.3 Model A Eval Result

0.3.1 Summary Metrics

row_count	question_answering_correctness/mean	\			
0	3.0	0.666667			
	question_answering_correctness/std	rouge/mean	rouge/std	bleu/mean	\
0		0.57735	0.78338	0.206721	0.62688
	bleu/std	exact_match/mean	exact_match/std		
0	0.356732	0.333333	0.57735		

0.3.2 Row-based Metrics

	prompt	\
0	Answer the question: Which part of the brain d...	
1	Answer the question: What provided the Roman s...	
2	Answer the question: What area did the Hasan-j...	
	response	\
0	frontal lobe and the parietal lobe	

```

1 The Roman Senate was filled with exuberance du...
2 The Hasan-Jalalians commanded the area of Syun...

```

```

                                reference \
0          frontal lobe and the parietal lobe
1          Due to successes against Catiline.
2 The Hasan-Jalalians commanded the area of Arts...

```

```

        question_answering_correctness/explanation \
0 The AI-generated response contains all claims ...
1 The AI response correctly states that the Roma...
2 The AI response states that "The Hasan-Jalalia...

```

```

        question_answering_correctness/score  rouge/score  bleu/score \
0                                1.0      1.000000    1.000000
1                                1.0      0.588235    0.289179
2                                0.0      0.761905    0.591460

```

```

        exact_match/score
0          1.0
1          0.0
2          0.0

```

0.4 Model B Eval Result

0.4.1 Summary Metrics

```

        row_count  question_answering_correctness/mean \
0          3.0                                0.0

        question_answering_correctness/std  rouge/mean  rouge/std  bleu/mean \
0                                0.0    0.231481    0.2228    0.04113

        bleu/std  exact_match/mean  exact_match/std
0 0.043073          0.0          0.0

```

0.4.2 Row-based Metrics

```

                                prompt \
0 Answer the question: Which part of the brain d...
1 Answer the question: What provided the Roman s...
2 Answer the question: What area did the Hasan-j...

```

```

                                response \
0                                Occipital lobe
1 The Roman Senate was subdued because they had ...
2 The Galactic Empire commanded the state of Utah.

```

```

                                reference \

```

```

0         frontal lobe and the parietal lobe
1         Due to successes against Catiline.
2 The Hasan-Jalalians commanded the area of Arts...

```

```

        question_answering_correctness/explanation \
0 The response is incorrect because it provides ...
1 The AI response states that the Roman Senate w...
2 The AI response does not contain any claims fr...

```

	question_answering_correctness/score	rouge/score	bleu/score	\
0	0.0	0.250000	0.000000	
1	0.0	0.000000	0.037478	
2	0.0	0.444444	0.085913	

	exact_match/score
0	0.0
1	0.0
2	0.0

```

[34]: referenced_eval_results = []
referenced_eval_results.append(("Model A", referenced_result_rag_a))
referenced_eval_results.append(("Model B", referenced_result_rag_b))

```

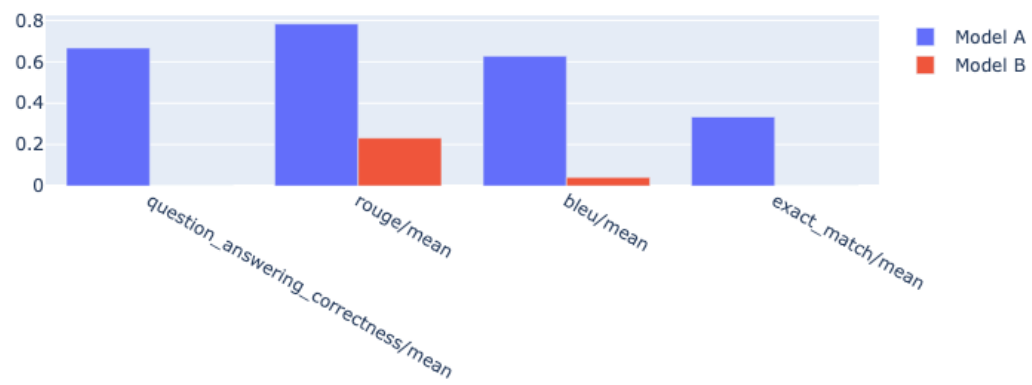
```

[35]: notebook_utils.display_radar_plot(
    referenced_eval_results,
    metrics=[
        "question_answering_correctness",
        "rouge",
        "bleu",
        "exact_match",
    ],
)

```



```
[36]: notebook_utils.display_bar_plot(  
    referenced_eval_results,  
    metrics=[  
        "question_answering_correctness",  
        "rouge",  
        "bleu",  
        "exact_match",  
    ],  
)
```



```
[ ]:
```